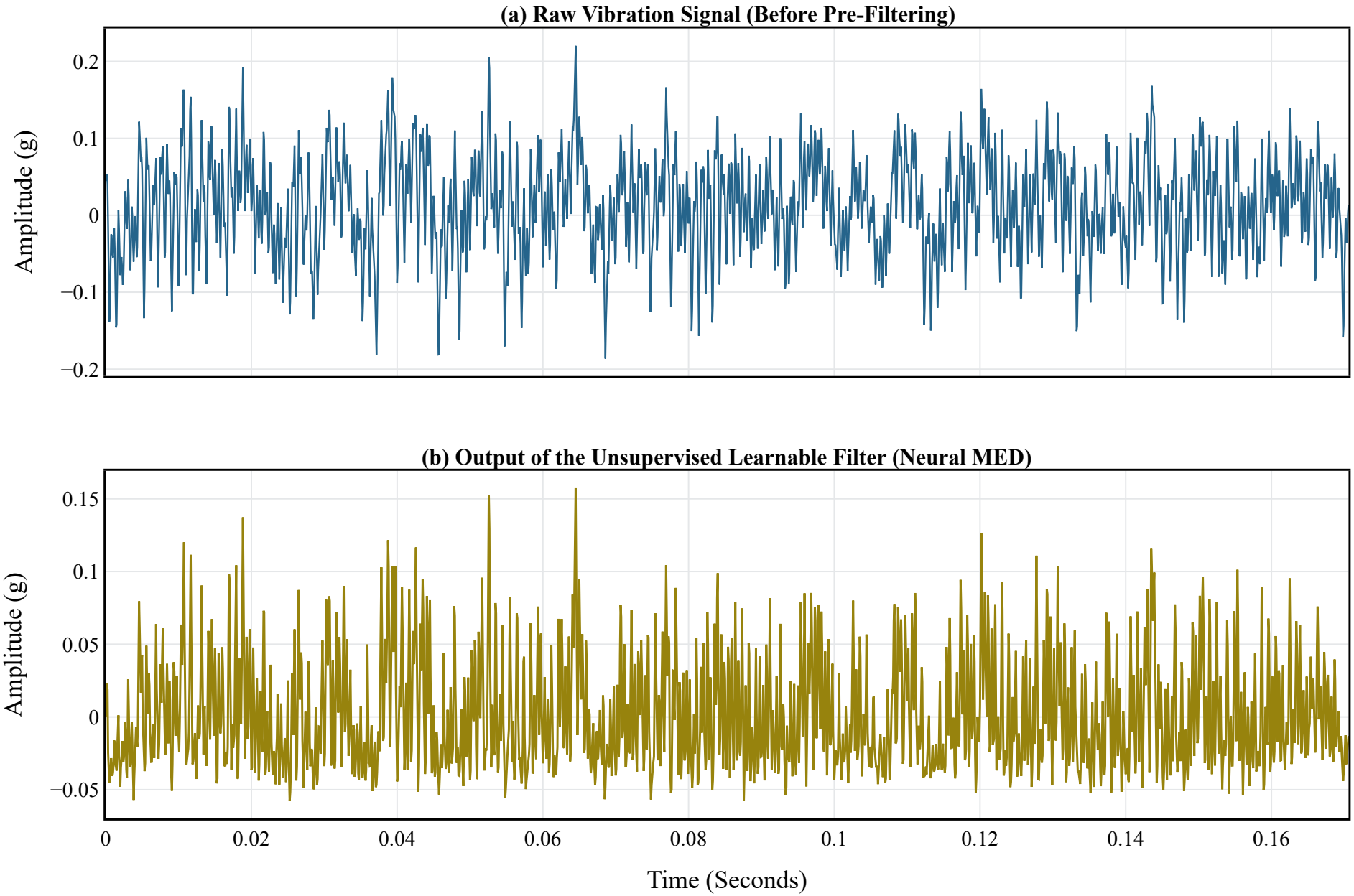

(a) Raw Vibration Signal (Before Pre-Filtering)
Amplitude (g)
0.2
0.1
0
−0.1
−0.2
(b) Output of the Unsupervised Learnable Filter (Neural MED)
Amplitude (g)
0.15
0.1
0.05
0
−0.05
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
Time (Seconds)